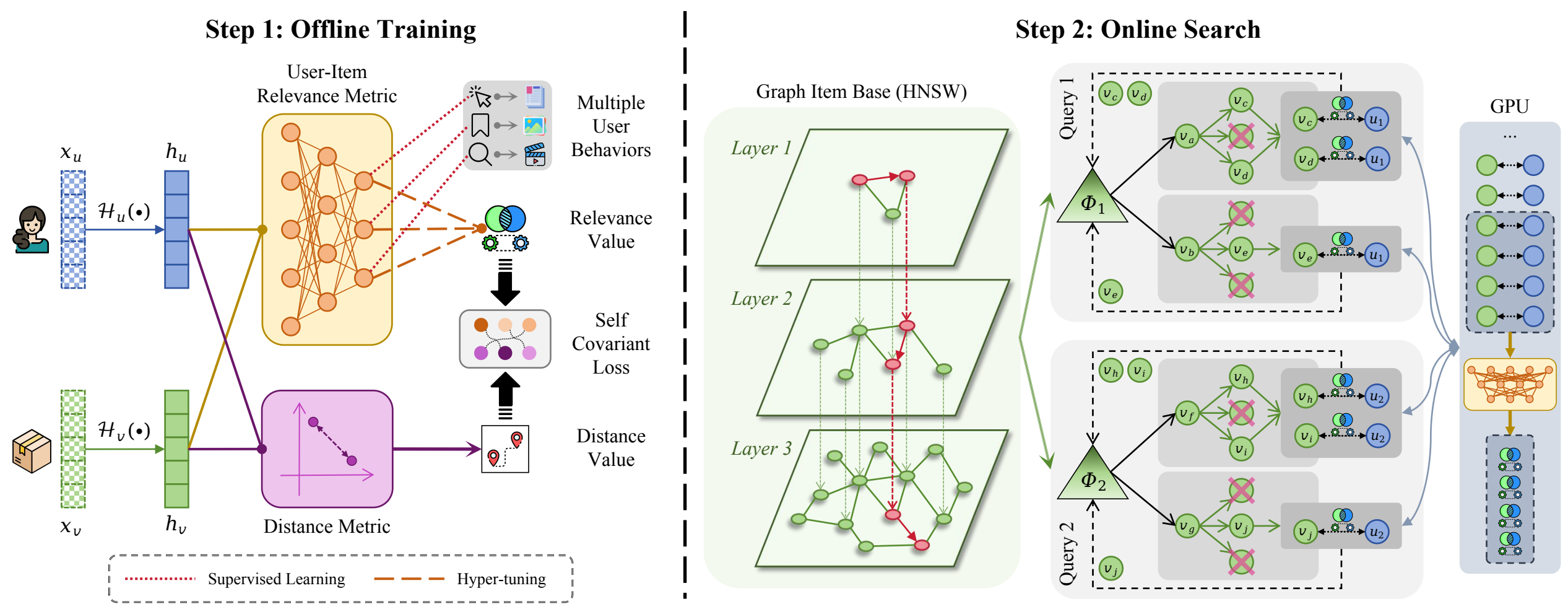
Step 1: Offline Training
User-Item
Relevance Metric
Multiple
User
Behaviors
Relevance
Value
Self
Covariant
Loss
Distance
Value
Distance Metric
Supervised Learning
Hyper-tuning
Step 2: Online Search
Graph Item Base (HNSW)
Layer 1
Layer 2
Layer 3
Query 1
Query 2
GPU